

more than 1500 kms parallel to the coast of Dronning Maud Land. This mountain belt, known as Fimbulheimen is considered to be a part of the Circum East Antarctic Mobile Belt (Yoshida; 1995). The major mountain ranges included in Fimbulheimen are as shown in the Fig.1. The area described as approximately in the center and lies between the Wohlthat range to the east and Muhlig-Hoffmanfjella to the west.

The Wohlthat range on the east, is a Proterozoic high grade metamorphic terrain with associated complex metamorphic, migmatitic, magmatic and deformation history comparable with other mountain ranges of Fimbulheimen mountain belt. The area was first described by Ravich and Soloviev(1966) and was subsequently studied by Indian geologists. The Wohlthat region exposes high grade ortho and para gneisses in Humboldt, Petermann and Schirmacher area (Sengupta,1988; Pant,1991; Joshi *et al.*, 1991; Ravindra *et al.*, 1994 and

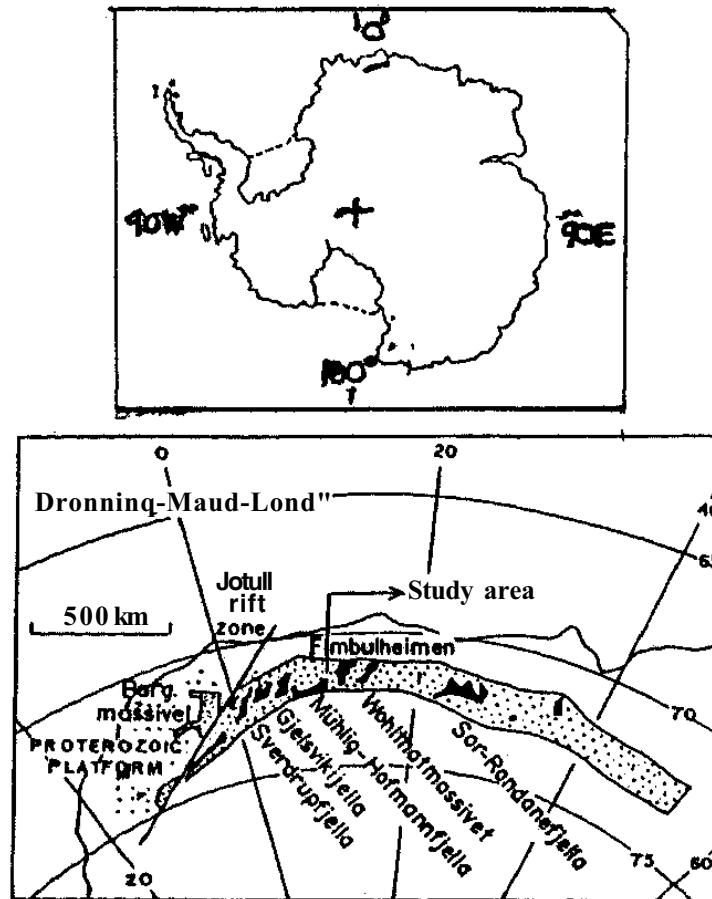


Fig1: General map of Dronning Maud Land with place names and major outcrops in Fimbulheimen are shown in black. The JP rift is also marked

D'Souza *et al.*, 1995) which reflect a peak metamorphism upto granulite facies with an overprint of amphibolite facies. The major manifestation of magmatic phase in Wohlthat range is in the form of a large Anorthosite massif in Gruber area (Ravich & Kamaney, 1975; Kaul *et al.*, 1991), foliated opx bearing granites (charnockites) from Skeids area, Charnockite suite rocks from Petermann ranges (Joshi & Pant; 1995), alkali granites from Petermann and Skeids area and charnockite-granite in intrusives from northern Humboldt (Ravindra & Kaul, 1995).

The Muhlig-Hoffmanfjella ranges on the west, mainly exposes metasediments comprising of migmatites, felsic gneisses and pelitic-semipelitic gneisses with intrusive gabbros (Ohta *et al.*, 1990; Bucher-Nurminen & Ohta, 1993). However the most dominant lithounits exposed in this region are the batholithic charnockites with some granites.

The geology of the other parts of Fimbulheimen i.e. Sverdrupfjella and Sor-Rondanfjella have been comprehensively described (Groenwald & Hunter, 1991; Allen, 1991 and Shiraishi *et al.*, 1991). The outcrops in Sor-Rondanfjella comprise of high grade metamorphic rocks, associated plutonic rocks and minor Mesozoic dykes. The H.U. Sverdrupfjella ranges expose mainly a sequence of para and orthogneiss and are suggestively products of regressive rehydration from granulite facies assemblages. Also occurring in the region are the younger alkaline plutonic complex and dolerite dykes related to extensional environment generated during the Gondwana rifting.

Geological Setting

On the basis of the field setting and petrographical characteristics, the rocks exposed in Dallmannfjellet area have been divided into three major units (Fig.2), as given below.

- (i) **Metamorphites:** Comprising layered heterogeneous garnet-biotite bearing quartzo-feldspathic gneisses, garnet-sillimanite-cordierite gneisses, granulite-amphibolite bands, garnetiferous pyroxene granulites and gneisses and bands of marble and calc-silicate rocks. Minor orthogneissic rocks of tonalitic to granitic composition also occur. Compositional heterogeneity exhibited through pronounced layering (Figs 3 and 4) is seen in the exposed rock at any given location. These rocks are dominantly exposed along the eastern ridge of the mapped area and also possibly as a restitic phase within the charnockite and granite.
- (ii) **Foliated charnockite:** It is exposed in the western outcrop of the area mapped and is closely associated with layered para and orthogneisses. The dominant foliation plane of the charnockite is visibly parallel to the

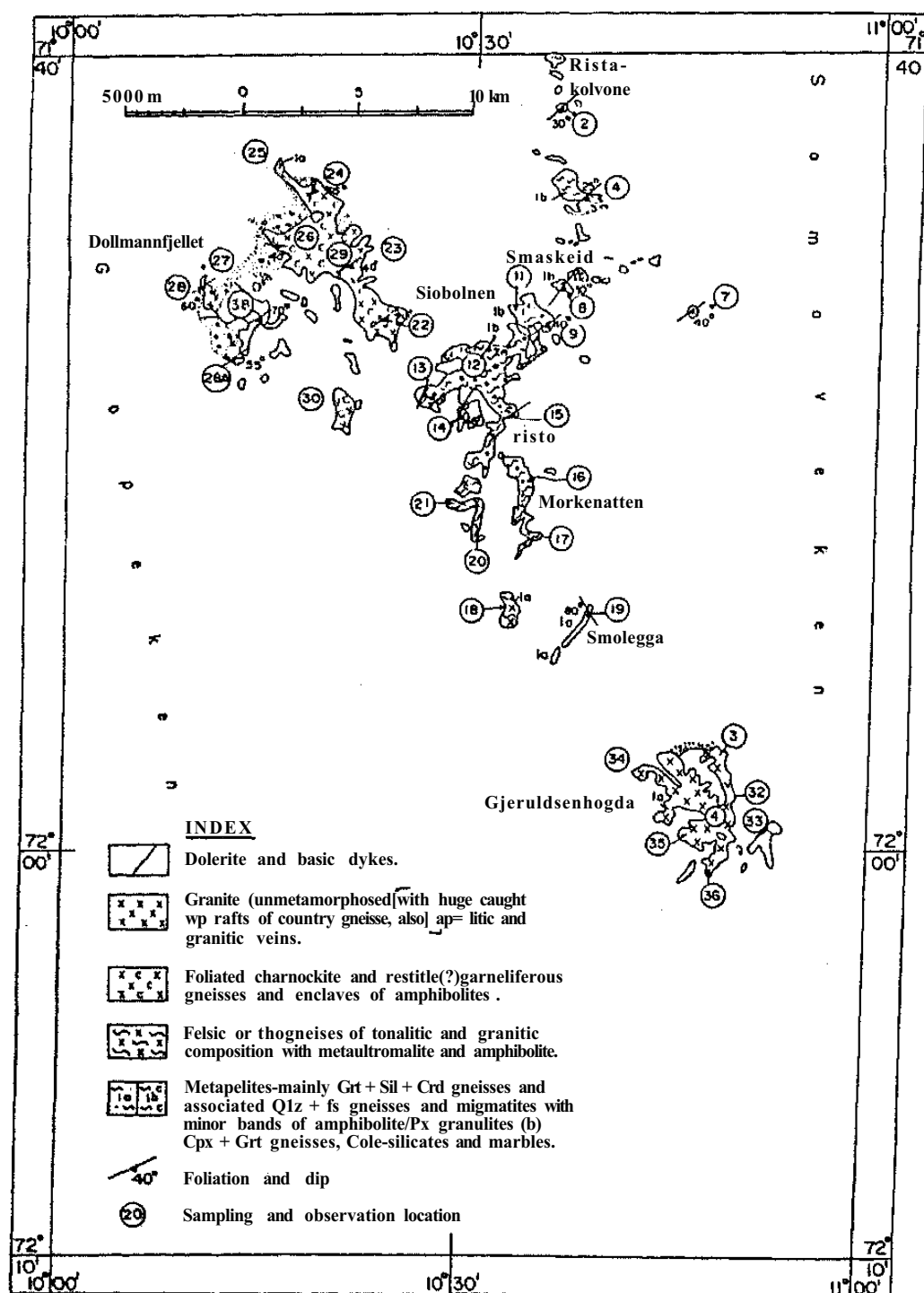


Fig.2: Geological map of the Dallmannfjellet area, CDML, East Antarctica



Fig.3: A view of the layered metamorphites in the background. The outcrop in foreground is of thinly foliated biotite gneisses traversed by quartz vein



Fig.4: An aerial view of the layered metamorphite showing compositional heterogeneity

regional penetrative gneissouty and is layered (Fig. 5). The foliated charnockite is intrusive into the country gneisses (Fig. 6). However at places the charnockite occurs as resistant bands within the layered gneisses and is parallel to the layering.

- (iii) **Porphyritic granite:** It is exposed in the entire southern outcrop. The granite is very coarse grained and contains megacrysts of alkali feldspar. Huge rafts of country gneisses (Fig.7) and restites of calc silicates (Fig.8) occur within the granite. Granite and aplitic veins intrude the foliated charnockite and country gneisses.

Besides the rock units described above, also occurring in the area in minor amount are quartz veins (sheared) and late basic dykes.

Structure

The layered metamorphites and foliated charnockites show distinct gneissosity. It is more pronounced in the paragneisses as it is defined by compositional layering and aligned mafic minerals. Preferred orientation of mafic minerals (mainly biotite) and at times stretched quartz marks the visible gneissosity in the foliated charnockite. In one sample (OR-23/1) of charnockite, differentiated layering is indicated by segregated biotite rich and biotite absent centimeter scale layers.

Discrete infrafolial isoclinal folds (F_1) present indicate the earliest deformational episode D_1 , forming the S_1 surface. The SF_1 surface has been folded into tight isoclinal folds (F_2) which are coaxial to F_1 folds. The mesoscopic F_2 folds observed in the area variably plunge towards SW. The D_2 deformation is however responsible for the pervasive regional gneissosity (S_2) which is almost NE-SW with variable southerly dip. A consistent L_2 mineral lineation with shallow plunge due north and south is seen developed in the layered gneisses probably indicating effect of D_3 deformation.

Ductile shear planes of more than one generation are present and have been manifested mainly in the form of layer parallel mylonites, reflecting syndeformational shearing. Faults associated with shearing were observed at two locations. The fault plane, trending $N20^\circ E-S20^\circ W$ has steep easterly dip. Pegmatite is seen intruding along this plane and the layered gneisses in proximity have developed drag folds.

Petrology

(1) On the basis of mineral assemblages, the metamorphites have been described. Compositionally, the metamorphites show prominence of metpelitic and semipelitic assemblages.



Fig.5. The outcrop of foliated clmrnockite, which appears as layered sequence. The foliated charnockite, as such reflects enderbitic to charnockite composition



Fig.6: The intrusive nature of foliated charnockite (lower) is highlighted. The rocks in upper part are the garnetiferous gneisses, and at the contact cuspal and lobate structure have developed. Also note that the penetrative fabric of upper unit is also penetrating within the lower parts



Fig.7: The outcrop of unmetamorphosed porphyritic granite with huge rafts of country gneisses (as marked)



Fig.8: The porphyritic granite with enclave of calc silicate rock

(a) Qtz+Kfs+Pl+Grt+Sil+RtCrd+Fe-Ti oxideBt+Zrn+Ap1a

Qtz+Kfs+Pl+Grt+Sil+Spl+Bt+Rt+Ap+Fe-Ti oxides. 2a

Qtz+Kfs+Grt+Bt+Fe-Ti oxide+Ap+Zrn. 3a

The mineral assemblages are akin to Khondalitic composition. The gneissosity is strongly reflected by ribbon quartz and aligned sillimanite, biotite etc. (Fig. 9). The intervening part shows granoblastic texture, which contains mainly quartzo-feldspathic minerals and stubby sillimanite. Perthitic and myrmekitic intergrowth are seen. The zircon present is mainly rounded (Fig. 10). Mylonitic fabric parallel to the main gneissic plane is present in some thin sections. Garnet invariably occurs as porphyroblasts and contain inclusions. Some of the inclusions present are (i) Poikiloblastic core with quartz inclusions (ii) swarms of sillimanite needles (Fig. 11) (iii) inclusions of coarse grained sillimanite, biotite, rutile and quartz. Reaction rims around garnet and sillimanite are prominently developed. Secondary mineralisation in the embayment of garnet and along the cracks is prominent.

(b) Qtz+Pl+Kfs+Cpx+Grt+Bt+Hbl+Bt+Zrn+Ap+Fe+Ti oxides.1b

Qtz+Kfs+Pl+Grt+Opx+Bt+Fe-Ti oxide+Cum. 2b

Light greyish coloured, thinly laminated and layered clinopyroxene bearing rocks are closely associated with quartzofeldspathic gneisses and sillimanite bearing metapelites. These rocks are in proximity of calc-silicate and marble bands. Development of a strong foliation with intervening granoblastic matrix is seen in the thin section. Porphyroblastic embayed garnet with reaction rim is present (Fig. 18). The garnet porphyroblast contains inclusions of quartz, plagioclase and biotite. Perthite and myrmekite intergrowth are prominent. Zircon occurs as rounded grains and elongated prisms. Hornblende and biotite content varies; hornblende, when present occurs as euhedral grains aligned along foliation. Mylonitic fabric developed parallel to foliation is seen (OR-4/5), wherein the clinopyroxene is porphyroblastic along with garnet.

Gametiferous orthopyroxene bearing gneiss (2b) too is a greyish coloured strongly foliated rock. The mineral assemblage indicated is characterised by prominent development of perthite and myrmekite. The orthopyroxene shows exsolved oxides along cracks and margin and also is altered to cummingtonite.

(c) Cpx+Scp+Wo+Pl+Ca+Qtz+Ep. 1c

The calc silicate bands occurring in close proximity of coarse grained white marble bands, in thin sections show essentially coarse grained granoblastic texture. Marginal symplectite has developed along the rim of scapolite. The calc silicate assemblage as a band is seen in a thin section of gametiferous quartzo-feldspathic gneiss (OR-8/3).

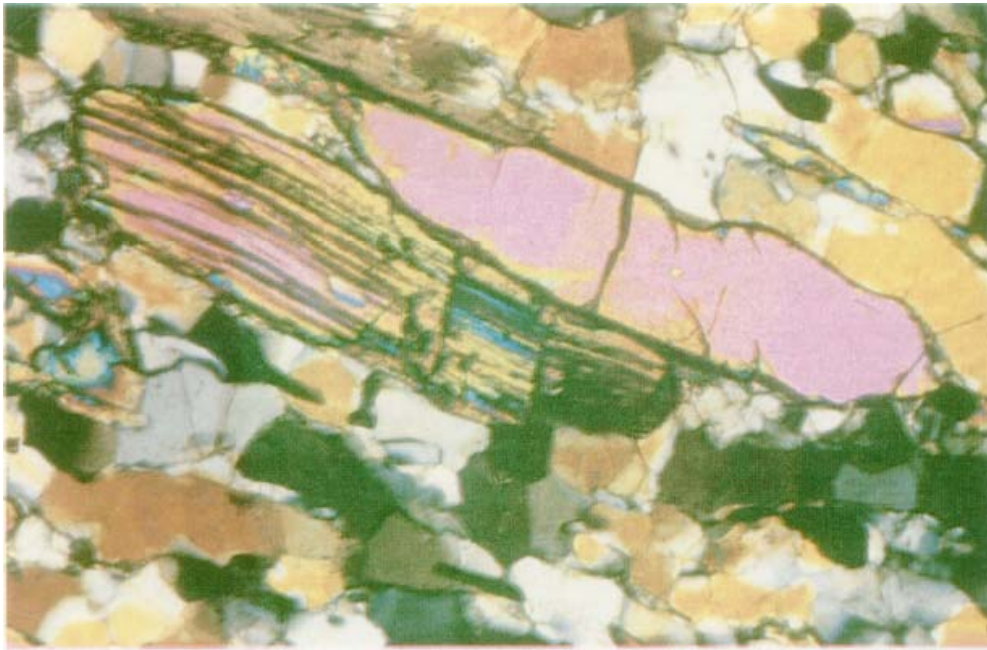


Fig.9: Photomicrography of sample no. OR-2/4 with asswmblag la. The Sillimanite, biotite and stretched quartz define the gneissocity (XP)



Fig.10: Photomicrograph of sample no. OR-14/2 with assemblage la. The rounded zircon with slight overgrowth highlight the paragneissic characteristics (XP)



Fig.11: Photomicrograph of sample no. OR-9/2 showing garnet porphyroblast with inclusion of sillimanite (as swarm) and quartz. The fractures developed have in them secondary biotite (XP)



Fig.12: Photomicrograph of sample no. OR-15/2 with assemblage 1a. The aligned stout sillimanite prisms are reflecting the original assemblage of Grt+Sil+Rt with Qtz and Kfs. The biotite parallel to the main foliation are developed during M2 metamorphism. The S_1 and S_2 surface is almost parallel (XP)

- (d) Pl+Cpx+Opx+Grt+Fe-Ti oxides+Hbl+BtSpn. Id
 Pl+Hbl+Cpx+Grt+Fe-Ti oxides+Bt+Spn. 2d
 Pl+Cpx+Hbl+Bt+Qtz+Kfs+Zrn+Ap. 3d
 Pl+Hbl+Bt+Qtz+Zrn+Ap+Zm. 4d

The mineral assemblages given above pertain to the metabasic rocks occurring as bands and interleaved layers within the layered gneisses. The metabasites indicate apparent compositional variety. The assemblages Id and 2d show pronounced formation of symplectite texture around garnet. The assemblage no. 3d shows coarse grained granoblastic texture with an overprint of biotite foliation. Hornblende present is secondary. The assemblage no. 4d is of amphibolite with coarse grained granoblastic texture. Profuse development of hornblende and quartz symplectite is seen.

- (e) Qtz+Kfs+Pl+Bt+Fe-Ti oxide+Zrn+Ap. 1e
 Qtz+Pl+Kfs+Hbl+Bt+Zrn. 2e

The assemblages given above are of orthogneissic rocks occurring within the layered paragneisses. The granite gneiss (1e) typically depicts coarse grained gneissic texture with porphyroblasts of K-feldspar and aligned elongated quartz. The orthogneiss contains an elongated enclave of meta ultramafite (OR-12/3), essentially composed of Opx+Cpx and Hb. The Zircons are occurring as square euhedral grains.

The assemblage 2e pertains to the orthogneiss with tonalitic bulk composition. This rock occurs within the layered gneiss and also within the foliated charnockite. It shows coarse grained gneissic texture with development of myrmekite and antiperthite.

(ii) The foliated charnockite exposed in the area exhibits two distinct mineral assemblages.

- (f) Qtz+Kfs+Pl+Opx+Cpx+Hbl+Bt+Fe-Ti oxide+Ap+Zrn. If

This rock is yellowish coloured gneissic and is closely associated with grayish coloured enderbitic charnockite. In this section, the gneissosity is well defined by elongated quartz and aligned mafics. The overall texture is coarse grained granular. Pethite (Micro & Meso) is prominent. The modal content of plagioclase and k-feldspar is almost equal. Alteration in orthopyroxene and clinopyroxene is seen. The orthopyroxene shows exsolved oxide along cracks and margin.

- (g) Qtz+Pl+Kfs+Opx+Cpx+Bt+Fe-Ti oxide+Hbl+Ap+Zrn. Ig
 Pl+Qtz+Kfs+Opx+Cpx+Fe-Ti oxide+Bt+Ap+Zrn. 2g

The mineral assemblage reflects enderbitic composition. These grayish coloured rocks in thin section show primary medium to coarse grained granular texture. The mafics present define the gneissosity. Hornblende and biotite in the rock appear to be of secondary origin. The orthopyroxene is altered with exsolved oxide along cracks and margins.

(iii) The granite exposed in the area is very coarse grained porphyritic and is composed of

Qtz+Kfs+Pl+Hbl+Bt+Fe-Ti oxide+Ap. 1h

Under the microscope it is coarse grained and depicts anhedral to subhedral inequigranular texture. Myrmekitic and perthitic intergrowth has prominently developed. Interstitial mafics and Fe-Ti oxide is present. The K-feldspar occurs as megacrysts. In very close proximity of granite a small stock of norite is exposed with mineral assemblage of

Opx+Pl+Kfs+Hbl+Bt+Fe-Ti oxide. 2h

and is characterised by subhedral granular texture. Orthopyroxene occurs as megacryst with inclusion of plagioclase and Fe-Ti oxides.

Metamorphism

Preliminary documentation of metamorphism and its relation with micro-textural reflection of deformation is done on the basis of detailed microscopic studies. Geothermometry and barometry are being done separately.

Polymetamorphism and deformation are indicated by the rocks exposed in the area. High grade metamorphism M₁ was synchronous with deformation which has produced S_j foliation. Aligned sillimanite with granoblastic quartz-feldspar and garnet represent the original assemblage and S_j surface (Fig. 12). The trail of sillimanite in porphyroblasts of garnet (Fig. 13) also aligned parallel to S_j surface. In section no. OR-2/7, sillimanite grain is folded with development of axial plane cleavage, along which biotite is present. The S_i surface is parallel to the main S_j surface. The S₂ surface is characterised by stretched ribbon quartz, sillimanite and flaky biotite (Fig. 14). The S₂ surface is also accenuated by mylonitic fabric. The S₃ surface present is indicated by biotite flakes developed oblique to the gneissosity and parallel to the broad fractures in garnet along which biotite is formed. The charnockite in Fig. 6 is also characterised by foliation plane parallel to the gneissosity (S₂) of the metamorphites. This foliation plane is mainly defined by aligned mafics and also by stretched quartz (Fig. 15), suggesting syn to pre M₂ intrusion.

The peak metamorphic assemblage present in thin sections studied are

(i) Grt+Sil+Crd+Rt+ilmBt

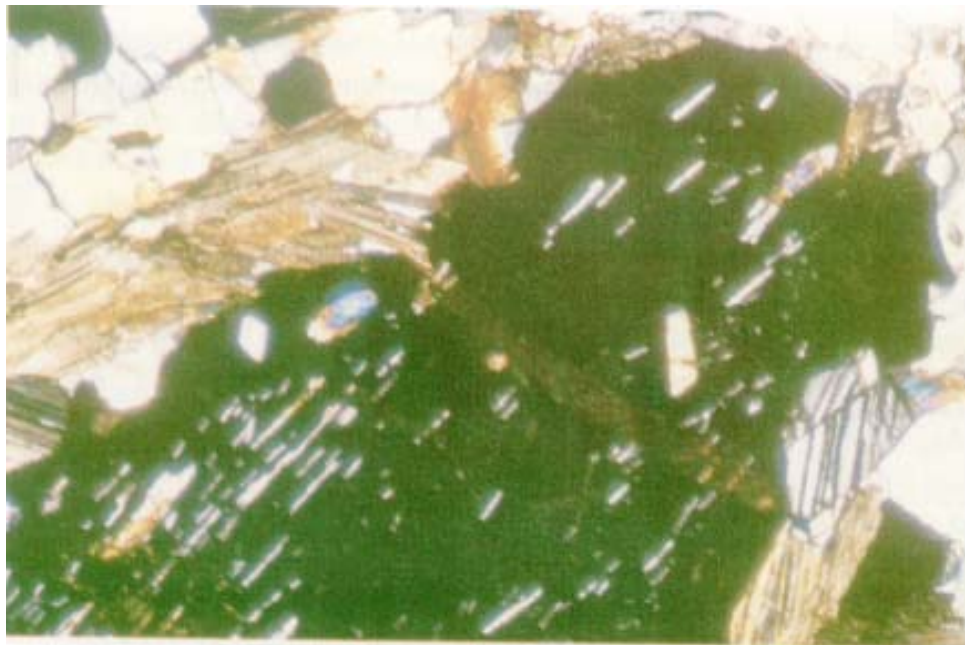


Fig. 13: Photomicrograph of sample no. OR-2/4 with garnet porphyroblast containing aligned sillimcmite as inclusion marking the S\ surface. Note the pervasive S2 fabric which is warping around the garnet porphyroblast is almost parallel to the S\ surface. The oblique fracture contains minute bio lite and probably confirms to S3 surface. Also seen Li tkwlopmmt oj'secmdmy bwitie « emboymmt (KPi



Fig. 14: Photomicrograph of sample no. OR-11/2 with assemblage la. The fibrolitic siUimanite is developed closely associated with hiotite and is different from coarse grained siUimanite, probably differentiating S\ and SJ surface (XP)

(ii) Grt+Sil+Spl+Rt

(iii) Grt+Opx

with quartz and feldspars.

The mineral paragenesis constrains the pressure-temperature conditions and indicates intermediate pressure granulite facies metamorphism. However, some of the metamorphic reactions present offer a rough estimation of P-T conditions attained during peak metamorphism.

The coexisting garnet sillimanite, Fe-Ti oxide (Ilmenite) rutile and quartz (Fig. 16) is indicative of a standard geothermometer commonly used (Bohlen *et al.*, 1983; Essene, 1989).

Grt+Rt ilm+Sil+Qtz (GRAIL)(1)

The most stable assemblage present in metapelitis however is Grt+Crd+SilBt+Kfs+Qtz, which may have progressed through following reactions (Bucher-Nurminen & Ohta, 1993), and explaining the presence of segregated garnetiferous leucogneiss layers.

Bt+Sil+Qtz Crd+Kfs+H₂O.(2)

Bt+Sil+Qtz Grt+Kfs+H₂O.(3)

Bt+Crd+Qtz Grt+Kfs+H₂O(4)

The limits Pressure-temperature during peak metamorphism thus can be roughly estimated between 6 to 7.5 kb of pressure and 700- 800°C Temperature (Schenk, 1994; Sheraton *et al.*, 1993; Waters, 1988). However the coexisting garnet and orthopyroxene may be reflecting prevailing higher pressure conditions during metamorphism.

Decompression and retrogressive rehydration reactions are common and are expressed as corona textures and reaction rim around garnet and sillimanite (Figs 17 & 18). The decompression indicated may be related to M2 metamorphism and D2 deformation. The cordierite rim between garnet and sillimanite (Fig. 19) is essentially a decompression reaction as given below (Harley & Fitzsimons, 1991).

Grt+SH+Qtz-*Crd.(5)

The formation of biotite & cordierite, biotite & quartz and biotite & plagioclase symplectites in garnet embayments (Fig.20) and formation of secondary sillimanite & biotite with quartz at the margin of garnet (Fig.21) indicate retrogressive rehydration reactions (6, 7, 8) as given below (Rasse & Schenk, 1994).

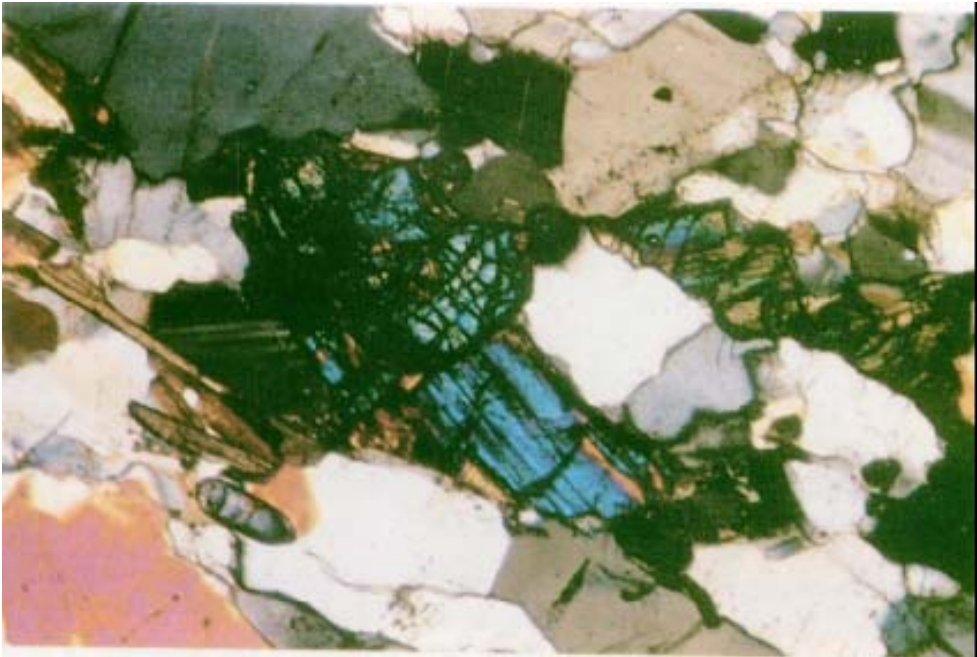


Fig. 15: Photomicrograph of sample no. OR-38/2. The gneissosity in the foliated charnockite is defined by aligned mafics and stretched quartz, confirming to S_2 surface of metamorphites (XP)

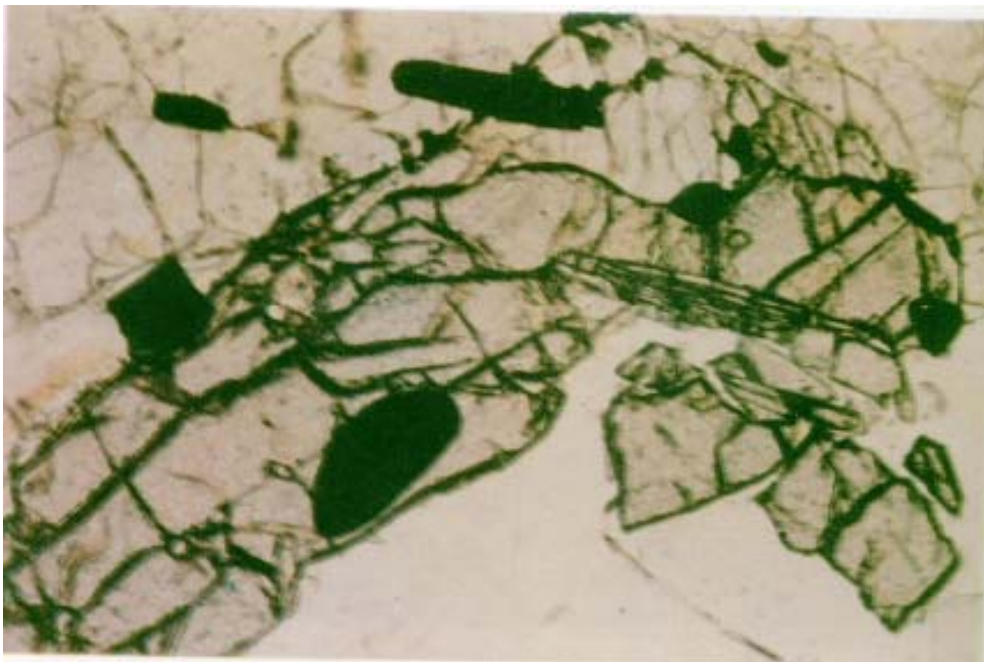


Fig. 16; Photomicrograph of sample no. OR-9/5 showing coexisting garnet, rutile, sillimanite and ilmenite (PP)



The biotite-sillimanite-quartz intergrowth in garnet embayment has been expressed by Schenk, (1984) as



Garnet breakdown to biotite and quartz along fractures confirming to S3 surface indicate increased influence of PH₂O during M3 metamorphism. The alteration of K-feldspars and muscovite in some section also indicate M3 related retrogression. Retrogressive replacement of orthopyroxene by biotite and quartz/ due to hydration is also seen in foliated chamockite. The flakey biotite defining the main S-surface in chamockite, however is related to M2 metamorphism.

Sample nos, OR/2/3 and 2/5 are of metabasic rocks closely interleaved with metapelites. These rocks show spectacular decompression reaction involving garnet with formation of orthopyroxene and plagioclase symplectite at the rim (Figs 22 a, b, c). Similar reactions have been described from East Antarctica (Hailey, 1988; Thost *et al.*, 1991) and from other granulite terrain (Schenk, 1984; Sandiford *et al.*, 1988). Harley (1988) has expressed it as dehydration reaction.



The symplectite formed as a result of reaction involving amphibole explains the presence of opaques.

The calc-silicate bands occurring within the metamorphites also show a retrogressive phase as indicated by the symplectite rim present around scapolite (Fig. 23). The formation of plagioclase, calcite and quartz symplectite can be explained by following reaction (Schenk, 1984).



The foliated chamockite also shows daetylitic intergrowth of quartz and biotite replacing orthopyroxene, suggesting regressive rehydration during M3 metamorphism.

Geochemistry

Whole rock analysis and normative composition of foliated chamockite and porphyritic granite is given in Table 1. The major oxides given were analysed by PW 1410 (PHILIPS) X-ray spectrometry system at PPOD laboratories, AMSE, G.S.I. Bangalore.

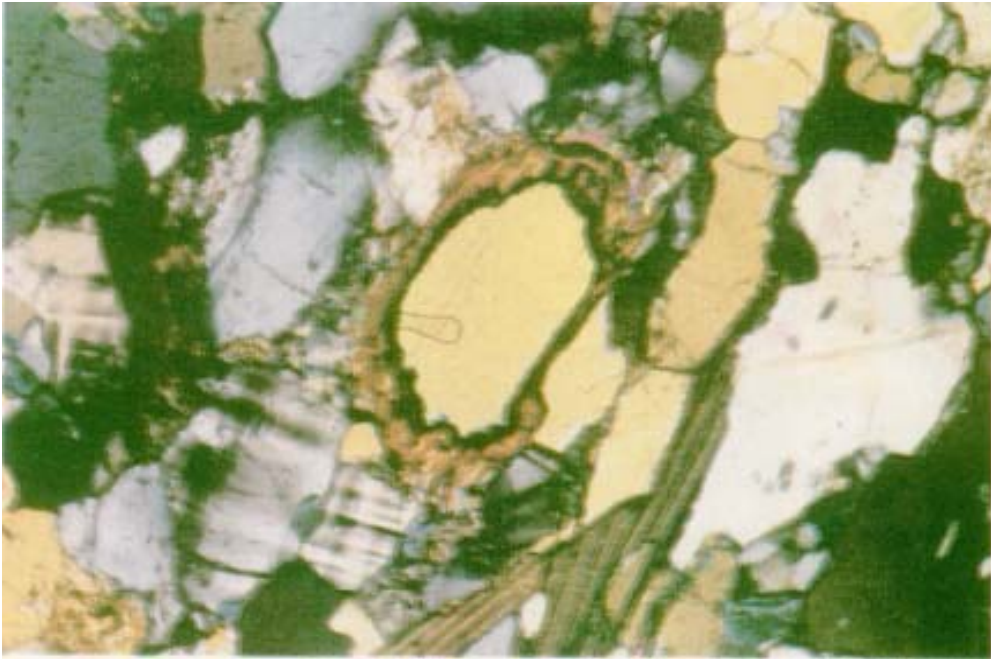


Fig. 17: Photomicrograph of sample no. OR-14/2 having $Grt+Sil=Crd$ assemblage. The sillimanite is seen with reaction corona (XP)



Fig. 18: Photomicrograph of garnet porphyroblast with reaction rim defined by symplectitic biotite, quartz and plagioclase. The rock is with assemblage ld (XP)

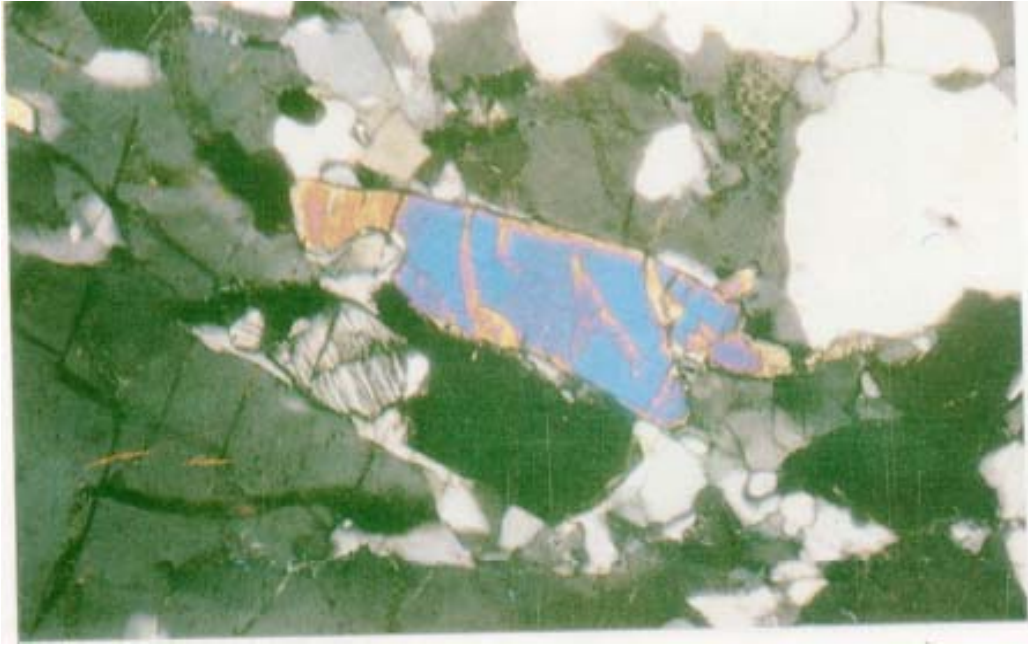


Fig .19: Photomicrograph of Grt-Sil bearing metapelite (OR-9/2). The reaction between garnet and sillimanite is indicated with formation of cordierite in between (XP)

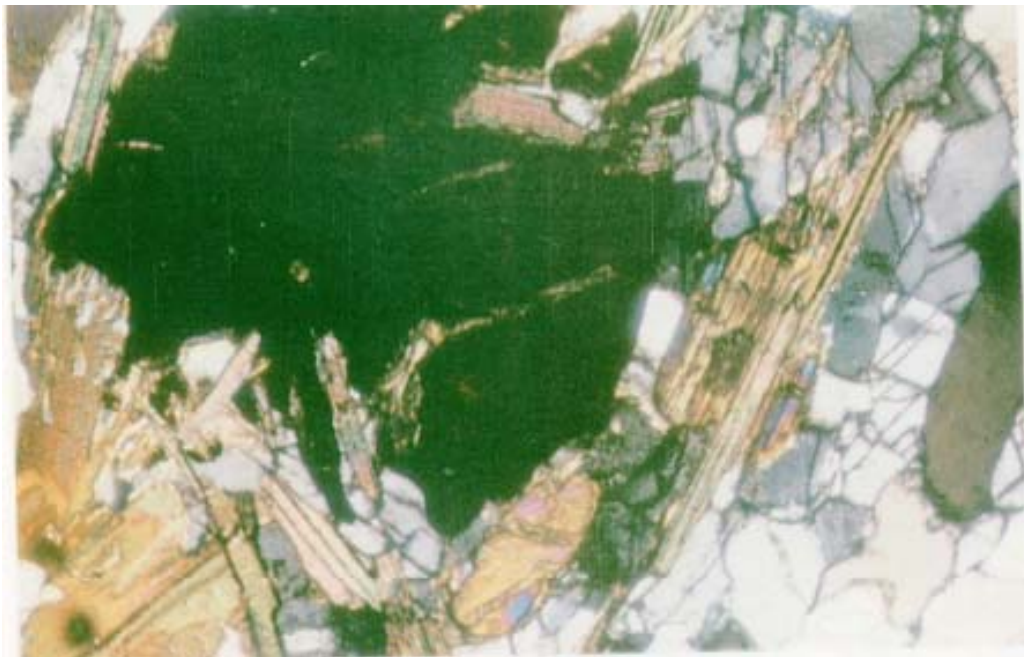


Fig.,20: Photomicrograph of sample no. OR-14-4 of garnet porphyroblast with biotite, sillimanite & cordierite. Secondary biotite due to rehydration is seen forming into garnet embayment. also present is cordierite between sillimanite and garnet (XP)

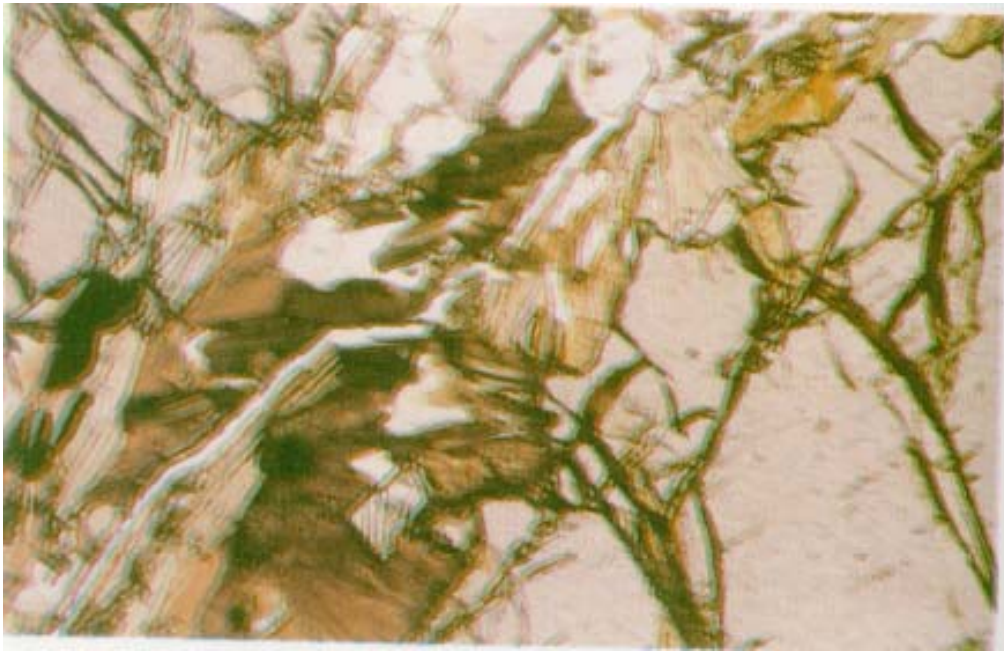


Fig. 21: Photomicrograph of garnet porphyroblast with included sillimanite and rutile (OR-2/7). Secondary sillimanite and biotite is formed at the margin. Also present in the embayment is symplectite of biotite and cordierite (PP)

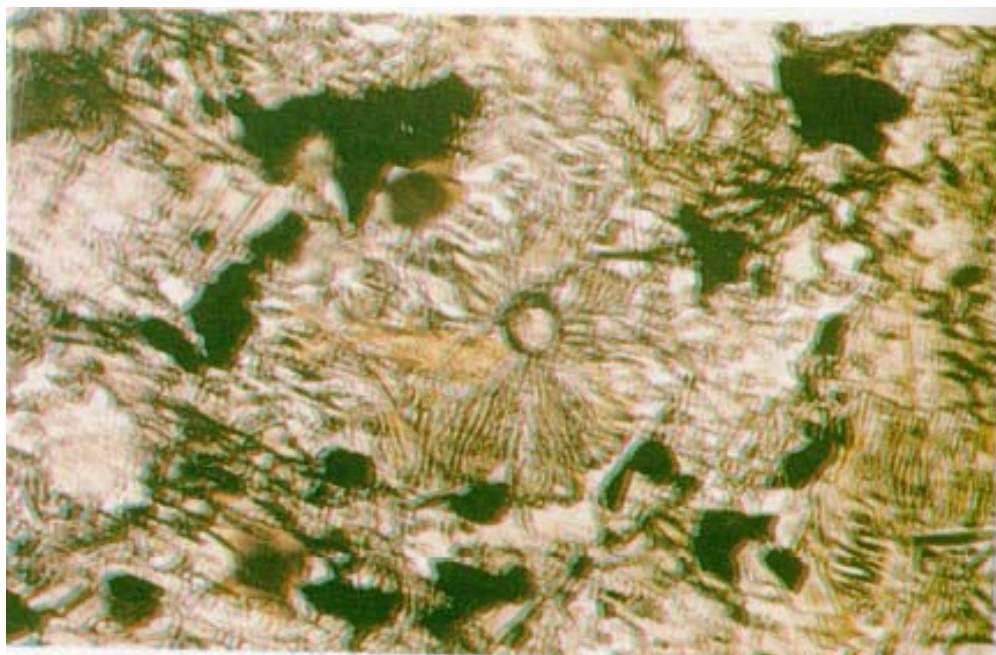


Fig.22a

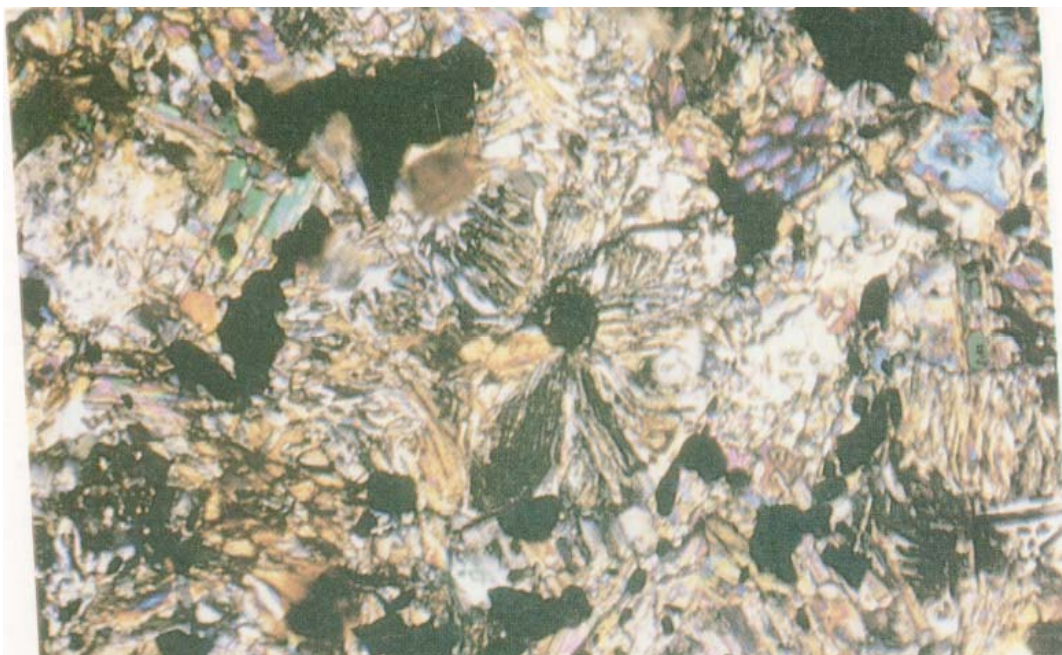
*Fig.22b*

Fig.22c: Photomicrograph showing isothermal decompression reaction (a&b) sample no. OR-2/3 in under PP & XP light. It shows a garnet grain surrounded by symplection of opx and plagioclase with hornblende. also present in the view are opaque (magnetite), clinopyroxene, hornblende and biotite (c) sample no. OR-2/6 with similar garnet breakdown reaction (XP)

Table 1: Whole rock analysis and normative values of porphyritic granite and foliated charnockite

Sample	OR35/1*	OR3675*	OR3674*	OR36/1*	OR36/3*	OR35/2*	OR27/1	OR27/2	OR27/4	OR28/2	OR28/2	OR38/2
SiO ₂	63.66	66.09	73.80	75.05	75.32	64.58	70.44	67.63	71.99	66.93	60.31	69.51
TiO ₂	1.08	1.04	0.24	0.27	0.31	0.93	0.35	0.72	0.50	0.94	2.38	0.52
Al ₂ O ₃	17.18	14.52	13.88	13.57	13.11	17.15	16.69	14.39	14.15	14.47	14.04	14.61
FeO	6.43	7.04	2.45	1.60	1.74	5.60	1.96	5.69	3.17	6.21	9.79	5.23
MnO	0.11	0.10	0.03	0.03	0.02	0.10	0.04	0.19	0.05	0.18	0.30	0.43
MgO	0.32	0.14	0.00	0.00	0.00	0.00	0.07	0.00	0.00	2.34	2.52	0.91
CaO	3.46	2.72	1.55	1.35	1.30	3.13	3.20	2.15	2.14	5.08	5.02	3.82
Na ₂ O	3.15	2.57	3.05	2.69	2.63	2.09	5.77	4.27	3.02	3.06	3.30	4.03
K ₂ O	4.26	5.45	4.99	5.43	5.55	5.14	1.42	4.86	4.82	0.62	1.70	0.80
P ₂ O ₅	0.34	0.34	0.02	0.02	0.02	0.26	0.06	0.10	0.15	0.16	0.66	0.13
Total	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
crpw**												
AN	35.91	34.23	22.71	22.32	22.11	34.55	24.05	13.71	27.38	47.99	39.82	34.67
Q	20.18	22.52	32.53	34.87	35.33	19.37	23.78	18.75	30.75	30.60	19.70	30.98
or	24.94	31.68	29.19	31.68	32.62	29.90	8.27	28.49	28.49	3.60	9.81	4.61
ab	26.40	21.41	25.55	22.51	22.09	25.72	48.23	35.79	25.55	25.55	27.25	33.17
an	14.80	11.14	7.51	6.47	6.27	13.58	15.28	5.69	9.64	23.58	18.03	17.61
C	1.90	0.24	0.68	0.85	0.46	1.43	0.00	0.00	0.42	0.00	0.00	0.48
di	0.00	0.00	0.00	0.00	0.00	0.00	0.01	3.77	0.00	0.19	1.72	0.00
hy	4.55	5.31	1.88	1.06	1.11	2.78	1.45	2.04	2.18	9.67	10.04	6.51
mt	3.73	3.65	1.30	0.84	0.93	3.51	1.04	3.20	1.70	3.52	5.54	2.91
il	2.03	1.94	0.46	0.51	0.59	1.75	0.66	1.35	0.95	1.77	4.41	0.97
ap	0.79	0.76	0.05	0.05	0.05	0.60	0.14	0.23	0.35	0.37	1.48	0.30
FeO*	6.43	7.04	2.45	1.60	1.74	5.60	1.96	5.69	3.17	6.21	9.79	5.23
F/F+M	0.953	0.980	1.000	1.000	1.000	1.000	0.966	1.000	1.000	0.732	0.800	0.8161
den	2.45	2.44	2.34	2.33	2.33	2.43	2.37	2.41	2.36	2.48	2.56	2.43

* Analysis of porphyritic granite

**CIPW norms recalculated to 100% free of volatiles



Fig.23: Photomicrograph of vale silicate rock no. OR-23/1. The large scapolite grain present is with symplectite rim (XP)

The foliated charnockite and granite, when plotted on normative QAP diagram (Fig. 24) show a compositional variability. The granites plot in syenogranite-Qz syenite-Qz-monzonite field, while the charnockites show a much wider spread. The spread in charnockites as discussed earlier is due to occurrence of co-existing charnockitic to enderbitic variety. The foliated charnockite in AFM diagram (Fig. 25) indicates a tholeiitic trend while no distinction is available for granite as it indicates only alkali enrichment.

In Molar $Ab_{0.3}/Na_{2O}+K_{2O}+CaO$ Vs. SiO_2 diagram (Fig. 26) the rocks are meta-aluminous and plot in igneous field. The A.S.I., in foliated charnockite decreases with decrease in SiO_2 content.

Discussions and Conclusions

The geological history of the rocks exposed in Dallmannjället area can be commented upon on the basis of microtextural and metamorphic textures described. The observations will however have substantiating contribution when it will be supplemented by geothermometric and barometric studies (under preparation).

The metamorphic peak is apparently represented by garnet+sillimanite+Rutile+Ilmenite+Biotite+Spinel and garnet+orthopyroxene assemblages. This indicates granulite facies metamorphism at midcrustal level. The peak

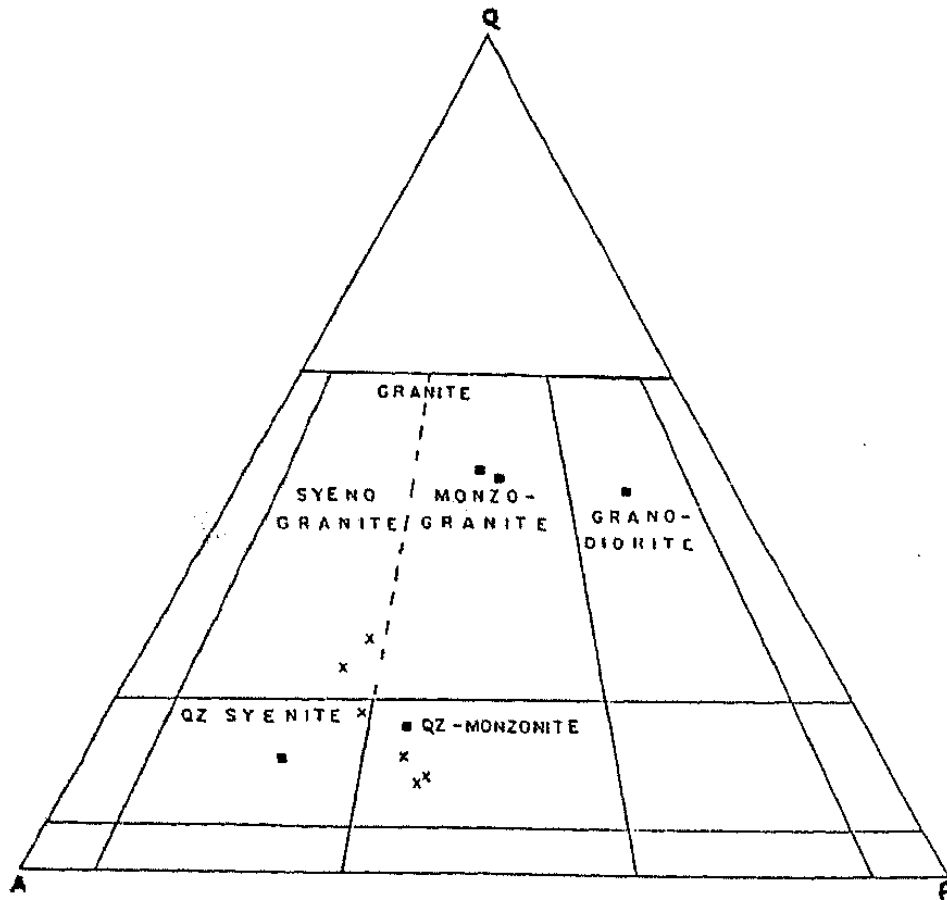


Fig.24: CIPW normative mineral contents of foliated charnockite (■) and porphyritic granite (x) plotted on the QAP diagram of Streckeisen (from Le Maitre, 1989)

metamorphic assemblages (M₁) have suffered primarily a decompression[^] reterogression (Reaction 5) during M₂ metamorphism which was associated with Dideformation and ductile shearing. The M₃ metamorphism is possibly reflected by the regressive rehydration reactions (6,7,8) under increased influence of PH₂O and represents the amphibolite fades overprint on granulite fades.

The strong decompression reaction (9) indicated by the metabasic records reterogression at a higher P-T condition (Thost *et al*, 1991) than the metamorphism reflected by the associated metapelites. It is to be ascertained if these mafic layers have shared a common history with the metamorphites or it provides an evidence for an earlier metamorphic event not preserved in the metapelites.

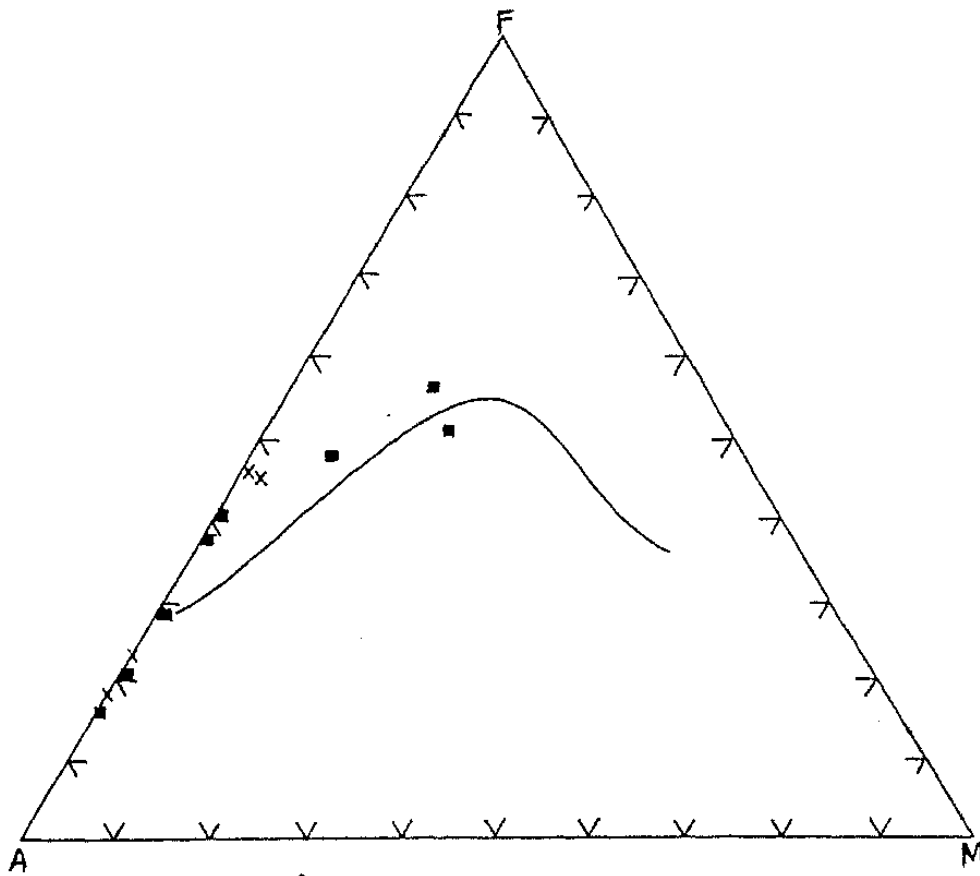


Fig.25: $\text{Na}_2\text{O} + \text{K}_2\text{O} - \text{FeO}$ - MgO (AFM) plot for foliated charnockite and porphyritic granite with dividing line between tholeiitic and calc alkeldine suite (After Irvine & Baragar, 1971)

The metamorphic evolution of the metamorphites can be best explained using a clockwise P-T trajectory path. The peak metamorphism (800°C temperature & 7.5 Kb pressure) must have been followed by decompression upto 5 kb pressure and finally cooling.

The foliated charnockites are clearly intrusive into the metamorphites (Fig. 6) and show strong igneous characteristics (to be published). The geological set up resembles the described characters of Mawson charnockite of E. Antarctica (Young and Ellis, 1991). The foliation in charnockite is concordant with the pervasive S2 fabric defined by metamorphites indicating probably syn to pre M2 intrusion. The foliated charnockite reflects a recrystallised texture, wherein the mafics (and at times stretched quartz) define the foliation, probably under the influence of M2 metamorphism.

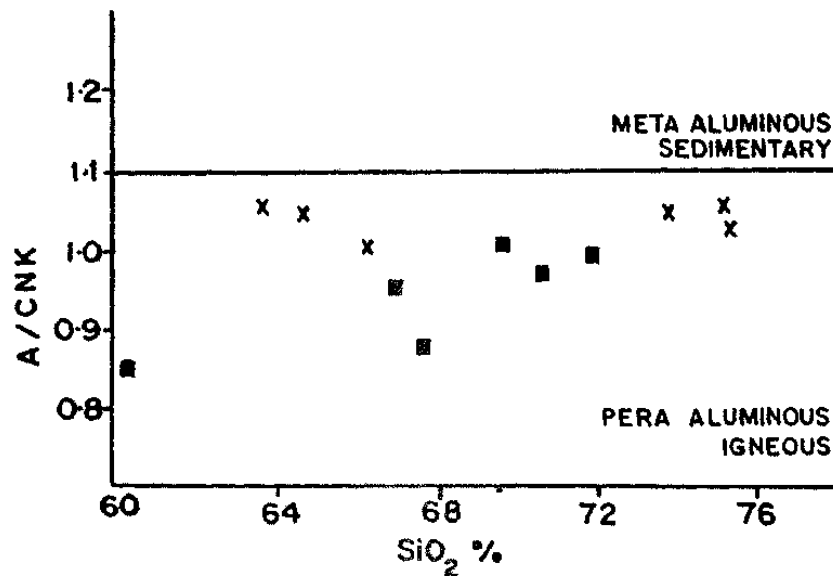


Fig.26: Plot of alkali-lime saturation index (A.S.I., molar $\text{Al}_2\text{O}_3/\text{CaO}+\text{Na}_2\text{O}+\text{K}_2\text{O}$) Vs SiO_2 for foliated charnockite and porphyritic granite (After White & Chappel, 1983)

The breakdown of orthopyroxene to biotite and quartz may reflect the M3 metamorphism. These foliated charnockites have been intruded by granite veins.

The porphyritic granite exposed shows no effect of metamorphism and contain huge rafts of country gneisses. The granite can be related to the Pan-African activity (500 Ma) which is reported in Dronning Maud Land.

The high grade metamorphic rocks in all probability are part of the proterozoic Circum East Antarctic Mobile belt (Yoshida, 1995) and have suffered peak granulite facies metamorphism at around 1000 Ma (Sheraton & Black, 1983; Black *et al.*, 1987; Groenwald & Hunter, 1991). The charnockite intrusion could be related with Mawson charnockite and represents pre to syn M2 magmatism at around 900 Ma (Young & Ellis, 1991).

Acknowledgements

The authors wish to thank D.G.; G.S.I, for nominating us to participate in XIII Antarctic Expedition and to Department of Ocean Development for providing all the logistics to complete the task. Shri M.K. Jha, Director, Antarctica Division is thanked for constant encouragement, guidance and facilities he provided during the expedition and afterwards to complete this paper.

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